Data Science & AI Mastery Syllabus

Phase 1: The Foundations (Weeks 1–6)

Module 1: Python Programming Fundamentals and Flow Control (Weeks 1-3)

- **Core Syntax & Variables:** Variable assignment, naming conventions, and basic math operators.
- **Primitive Data Types:** Integers (int), decimals (float), text (str), and booleans (True/False).
- **String Manipulation:** Slicing, methods (.upper(), .replace(), .split()), and f-strings.
- **Data Structures:** * Lists (ordered, mutable).
  - Dictionaries (key-value pairs).
  - Tuples (ordered, immutable).
  - Sets (unique values, intersections).
- **Control Flow (Logic):** Comparison/logical operators, and if, elif, else statements.
- **Loops:** for loops and while loops for iteration.
- **Functions:** def keyword, arguments/parameters, local vs. global scope, and return statements.

Module 2: The Data Handling Toolkit NumPy and Pandas (Weeks 4-6)

- **NumPy Basics:** Creating ndarrays, multidimensional arrays, and array broadcasting.
- **Pandas Core Objects:** Series (1D) vs. DataFrame (2D).
- **Importing/Inspecting Data:** pd.read_csv(), .head(), .info(), and .describe().
- **Data Cleaning:** Handling missing data with .isna(), .dropna(), and .fillna().
- **Data Manipulation:** Boolean indexing (filtering), aggregating with .groupby(), and combining datasets via .merge() and .concat().

Phase 2: The Analytical Mindset (Weeks 7–10)

Module 3: Statistical Thinking, Data Prep, and ML Workflow (Weeks 7-10)

- **Descriptive Statistics:** Mean, median, mode, variance, standard deviation, and percentiles.
- **Data Distribution:** Normal distributions vs. skewed data.
- **Data Visualization:** * Matplotlib (line charts, scatter plots, histograms).
  - Seaborn (boxplots, correlation heatmaps).
- **Feature Engineering/Scaling:** One-Hot Encoding for categories, Min-Max Normalization, and Z-score Standardization (StandardScaler).

- **The ML Pipeline:** Defining features (X) and targets (y).
- **Data Splitting:** Using train_test_split for training, validation, and testing sets.

Phase 3: Machine Learning Mastery (Weeks 11–20)

Module 4: Core Supervised Learning: Regression and Classification (Weeks 11-15)

- **Supervised Learning:** Regression (predicting numbers) vs. Classification (predicting categories).
- **Regression Algorithms:** Simple and Multiple Linear Regression math and intuition.
- **Regression Metrics:** Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared.
- **Classification Algorithms:** Logistic Regression (Sigmoid curve) and Decision Trees.
- **Classification Metrics:** Confusion Matrix, Accuracy, Precision, Recall, and F1-Score.

Module 5: Advanced ML concepts and Unsupervised Learning (Weeks 16-20)

- **Model Robustness:** Overfitting vs. Underfitting and the Bias-Variance Trade-off.
- **Cross-Validation:** K-Fold Cross Validation.
- **Ensemble Methods:** Bagging vs. Boosting, Random Forests, and Gradient Boosting (e.g., XGBoost).
- **Hyperparameter Tuning:** Grid Search (GridSearchCV).
- **Unsupervised Learning:** Working with unlabeled data.
- **Clustering:** K-Means algorithm, centroids, and the "Elbow Method."

Phase 4: The Modern AI Frontier (Weeks 21–26)

Module 6: Advanced Generative AI Concepts and Tools (Weeks 21-24)

- **Foundation Models:** LLM architecture basics and differences between GPT, Gemini, and Llama.
- **Prompt Engineering:** Zero-shot, Few-shot, Chain-of-Thought (CoT), and system instructions.
- **Embeddings & Vector Databases:** Text embeddings, Cosine Similarity, and Vector DBs (Chroma, Pinecone).
- **RAG (Retrieval-Augmented Generation):** Grounding LLMs in private documents.
- **AI Agents:** Tool-usage frameworks (LangChain, LlamaIndex) and multi-step planning.

Module 7: Capstone Project and Final Presentation (Weeks 25-26)

- **Project Scoping:** Translating a business problem into a data science problem.

- **End-to-End Execution:** Building the full pipeline from data sourcing to model evaluation.
- **Version Control:** Git/GitHub basics for portfolio hosting.
- **Storytelling with Data:** Presenting technical metrics as actionable business insights.